Information Security

Midterm – Pattern

Prepare lectures covered in slides 1,2,3

Total Marks: 30
Total Time: 1 hour

Q1. Conceptual Understanding (e.g. define Information Security along with its components)
Q2. Analytical Reasoning (e.g. Provide one practical example of a situation where improving one component of security might weaken another, and suggest how a balanced security strategy can be achieved.)
Q3. Apply the knowledge to solve (e.g., apply ceaser cipher algorithm on 'xyz' text for encryption and decryption)

Final Exam Pattern

Prepare lectures covered in slides

- IS - Slides 3 - Encryption.ppt
- IS - Slides 4 - Encryption - AES.ppt
- IS - Slides 4 - Further Details - Modular Arithmetic in Cryptography.pdf
- IS - Slides 4 - Further Details - Modular Arithmetic.pdf
- IS - Slides 5 - Hashing - More Details.ppt
- IS - Slides 5 - Hashing.ppt
- IS - Slides 6 - Authentication.ppt

Topics to Prepare:

- Network Security
- Cryptography, Definitions and Concepts,
- Cipher Methods (Bit Stream, Block Stream). Substitution Cipher (Monoalphabetic substitution, Polyalphabetic substitution, Vigenère cipher)
- Transposition Cipher (Caeser Matrix, Exclusive OR (XOR), Varnam Cipher, Book or Running Key Cipher)
- Symmetric Key Cryptography. Private Key Distribution Approaches. Stream Ciphers vs. Block Ciphers. DES Algorithm vs. AES Algorithm
- Asymmetric Key Cryptography. Public Key Distribution Approaches. RSA Algorithm vs. PGP Application. Comparison of Symmetric and Asymmetric Key Cryptography
- Intrusion Detection and Prevention Systems
- Signature based IDPS, Behavior Based IDPS, Network-based IDPS, Host-based IDPS

Types of Questions:

Q1. Theoretical- 10 Marks [15 minutes]
- 1.a
- 1.b

Q2. Theoretical – 10 Marks [15 minutes]
- 2.a
- 2.b

Q3. Mathematical – 15 marks {algorithm} [30 minutes]
Q4. Mathematical – 15 Marks [30 minutes]

Total Marks: 50
Total Time: 1 hour and 45 minutes